

# The Construction of Credit Curve Using Duffie-Singleton Approach

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## References

### 1 Introduction

This note presents a curve construction method based on Duffie and Singleton method. It has two parts: a discrete method and a fully continuous method. It shows how exactly marginal martingale default probabilities can be extracted using risky bonds.

A bondholder is entitled to the face value plus the accrual interest when a coupon bearing bond defaults. If we consider default only, an investor should buy discount bonds and sells premium bonds. However, this strategy is hard to realize a good profit as default is a rare event, it is hard to construct an credit default arbitrage portfolio. Another default treatment assumption is made by Jarrow & Turnbull [2], who assumes that bondholder can recover a fixed percentage of the present value of all remaining cash flows but valued at risk free interest rate curve. This is not realistic in practice as for a high yield bond with high coupon, its recovery value could be higher than the bond price before the default. The approach is based on a credit migration matrix. In practice, we do not use rating for valuation as rating is very much a classification method, which does not provide a granular valuation basis.

Duffie and Singleton [1] assumes that a bondholder is entitled to a fixed percentage of the bond price just before default. There is a concern that when a bond is about to default, it should be traded at a recovery value. However, the counter argument is that default is not a *predictable event*, you never know when default would occur. We use Duffie-Singleton approach to construct a credit curve based on cash instruments such as bonds or asset swaps. Another reason is that the credit curve constructed using this approach has a much smoother forward credit spreads even under the constant flat hazard rate assumption.

## 2 A Discrete Approach

Assume a bond promises a stream of cashflows  $C_1, C_2, \dots, C_n$  at future times  $t_1, t_2, \dots, t_n$ .

- $t_0$ : The current time.
- $D(t_0, t)$ : The risk-free discount factor (price of a zero-coupon Treasury bond paying \$1 at time  $t$ ).
- $V(t_i)$ : The bond's market value at time  $t_i$  immediately *after* cashflow  $C_i$  is paid.

Using Diffie-Singleton assumption, we have the following recursive formula for the valuation:

$$\begin{aligned} V(t_i) &= \frac{D(t_0, t_{i+1})}{D(t_0, t_i)} [p_i[C_{i+1} + V(t_{i+1})] + (1 - p_i)R[C_{i+1} + V(t_{i+1})]] \quad (1) \\ &= \frac{D(t_0, t_{i+1})}{D(t_0, t_i)} [p_i + (1 - p_i)R][C_{i+1} + V(t_{i+1})], \quad i = 0, 1, 2, \dots, n-1, \end{aligned}$$

and

$$V(t_n) = 0.$$

If we do consecutive substitution in (1) we obtain

$$V(t_0) = \sum_{i=1}^n D(t_0, t_i) \left( \prod_{j=0}^{i-1} [p_j + (1 - p_j)R] \right) C_i \quad (2)$$

If we denote

$$D^T(t_0, t_i) = D(t_0, t_i) \left( \prod_{j=0}^{i-1} [p_j + (1 - p_j)R] \right) \quad (3)$$

we have

$$V(t_0) = \sum_{i=1}^n D^T(t_0, t_i) \cdot C_i.$$

We can say  $D^T(t_0, t_i)$  is the total discount factor which consists of two parts: risk free discount factor, and *credit discount factor*. This separation of the total discount factor due to interest rate and credit is valid in the deterministic assumptions of interest rates and hazard rates. By no arbitrage, the forward price for a zero coupon, default contingent bond over the interval  $[t_{i-1}, t_i]$  with payment at time  $t_i$  is

$$\frac{D(t_0, t_{i+1})}{D(t_0, t_i)} [p_{i-1} + (1 - p_{i-1})R].$$

From equation (3) we see this is exactly equal to  $\frac{D^T(t_0, t_{i+1})}{D^T(t_0, t_i)}$ . To extract the marginal default probability, then we can just use expression (2) and the following definition

$$p_i = e^{-\int_{t_i}^{t_{i+1}} h(s) ds}.$$

If we assume a piecewise constant hazard rate, that is

$$h(s) = h_i, \quad s \in (t_{i-1}, t_i],$$

then we have

$$p_i = e^{-h_{i+1}(t_{i+1}-t_i)}.$$

### 3 A Continuous Model

Using Duffie and Singleton approach we assume that the default payment is given by

$$X(t^-)(1 - L(t))$$

where

- $X(t^-)$  is the value of instrument just before default
- $L(t)$  is the loss function given default

Then, every instrument can be valued using a recursive formula as follows

$$X(t) = P(t)E^* \left[ \frac{X(T)}{P(T)} \Pr[\tau > T | \tau > t] + \int_t^T \frac{X(s)(1 - L(s))}{P(s)} f(s | \tau > t) ds \right] \quad (4)$$

Since

$$\Pr[\tau > T | \tau > t] = \frac{\Pr[\tau > T]}{\Pr[\tau > t]} = \frac{S(T)}{S(t)},$$

where  $S(t)$  is the survival function, and

$$f(s | \tau > t) = \frac{f(s)}{S(t)}.$$

We can rewrite equation (4) as follows

$$X(t) = P(t)E^* \left[ \frac{X(T)S(T)}{P(T)S(t)} + \int_t^T \frac{X(s)(1 - L(s))}{P(s)S(t)} f(s) ds \right].$$

Rearranging this equation we have

$$\frac{X(t)S(t)}{P(t)} = E^* \left[ \frac{X(T)S(T)}{P(T)} + \int_t^T \frac{X(s)(1 - L(s))}{P(s)} f(s) ds \right].$$

Denoting

$$\frac{X(t)S(t)}{P(t)} = x(t), \quad (5)$$

we have

$$\begin{aligned}
x(t) &= E^*[x(T) + \int_t^T x(s)(1 - L(s))f(s)/S(s)ds] \\
&= E^*[x(T) + \int_t^T x(s)(1 - L(s))h(s)ds] \\
&= E^*[x(T) + \int_0^T x(s)(1 - L(s))h(s)ds] - \\
&\quad \int_0^t x(s)(1 - L(s))\mu(s)ds]
\end{aligned}$$

which implies that

$$M(t) = x(t) + \int_0^t x(s)(1 - L(s))h(s)ds$$

is a martingale.

$$dx(t) = -x(t)(1 - L(t))h(t)dt + dM(t)$$

which can be solved as

$$x(t) = e^{-\int_0^t (1-L(s))h(s)ds} m(t),$$

$$m(t) = x(t)e^{\int_0^t (1-L(s))h(s)ds}$$

is still a martingale.

$$m(t) = E^*[m(T)]$$

we have

$$x(t)e^{\int_0^t (1-L(s))h(s)ds} = E^* \left[ x(T)e^{\int_0^T (1-L(s))h(s)ds} \right].$$

Substituting (5) into above equation, we have

$$\frac{X(t)e^{-\int_0^t h(s)ds}}{P(t)} e^{\int_0^t (1-L(s))h(s)ds} = E^* \left[ \frac{X(T)e^{-\int_0^T h(s)ds}}{P(T)} e^{\int_0^T (1-L(s))h(s)ds} \right],$$

that is

$$\frac{X(t)}{P(t)} = E^* \left[ \frac{X(T)}{P(T)} e^{-\int_t^T L(s)h(s)ds} \right].$$

If we assume that

$$P(t) = e^{-\int_0^t r(s)ds},$$

then we have

$$X(0) = E^*[X(T)e^{-\int_0^T [r(s)+L(s)h(s)]ds}],$$

which is the fundamental result in Duffie & Singleton.

This implies that under the equivalent martingale  $m(t)$ , a risk neutral investor of an instrument subject to default has a rate of return  $r(s) + L(s)h(s)$ , instead of the usual  $r(s)$  in the absence of default. If we have correlation assumption for the stochastic interest rate and hazard rate, then the total discount factor is not separable into two parts.

## References

- [1] Darrell Duffie and Kenneth J. Singleton, Modeling term structures of defaultable bonds, Review of Financial Studies, 1999, Vol 12, Num. 4, pp.687-720.
- [2] Robert A. Jarrow and Stuart M. Turnbull, Pricing Derivatives on Financial Securities Subject to Credit Risk, Journal of Finance, 1995, Vol. 50, Number 1, pp. 53-85.
- [3] David X. Li, Constructing a Credit Curve, Credit Risk, A Risk Special Report, November 1998, pp. 40-44.